

● MEDIUM

● **ADVANCED MATH**

Advanced Math

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

GRID-IN ADVANCED MATH

The function f is defined by $fx = 5\frac{1}{4} \cdot x^2 + \frac{11}{4}$. What is the value of $f\frac{1}{4}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

ADVANCED MATH

Which quadratic equation has exactly one distinct real solution?

- A. $x + 15^2 = 0$
- B. $x + 15^2 = -45$
- C. $x + 15^2 = 45$
- D. $x + 15^2 = 135$

Q3

ADVANCED MATH

If $x^2 = a + b$ and $y^2 = a + c$, which of the following is equal to $(x^2 - y^2)^2$?

- A. $a^2 - 2ac + c^2$
- B. $b^2 - 2bc + c^2$
- C. $4a^2 - 4abc + c^2$
- D. $4a^2 - 2abc + b^2c^2$

A physics class is planning an experiment about a toy rocket. The equation $y = -16x^2 - 5.6x + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- A. This toy rocket reaches an estimated maximum height of 502 feet 16 seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of 16 feet 502 seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of 5.6 feet 502 seconds after it is launched into the air.

$$p = \frac{k}{4j + 9}$$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

- A. $4j + 9 = \frac{k}{p}$
- B. $4j + 9 = kp$
- C. $4j + 9 = k - p$
- D. $4j + 9 = \frac{p}{k}$

$$f(x) = x^2 + bx$$

$$g(x) = 9x^2 - 27x$$

Functions f and g are given, and in function f , b is a constant. If

$f(x) \cdot g(x) = 9x^4 - 26x^3 - 3x^2$, what is the value of b ?

- A. -26
- B. $-\frac{26}{9}$
- C. $\frac{1}{9}$
- D. 9

$$f(x) = 4,000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after 2012, where $x \leq 7$. Which of the following is the best interpretation of 4,000 in this context?

- A. The estimated initial beetle population for this species and area in 2012
- B. The estimated beetle population for this species and area 7 years after 2012
- C. The estimated percent decrease in the beetle population for this species and area each year after 2012
- D. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

Q8

GRID-IN

ADVANCED MATH

$$y = x^2 - 4x + 4$$
$$y = 4 - x$$

If the ordered pair (x, y) satisfies the system of equations above, what is one possible value of x ?

STUDENT-PRODUCED RESPONSE

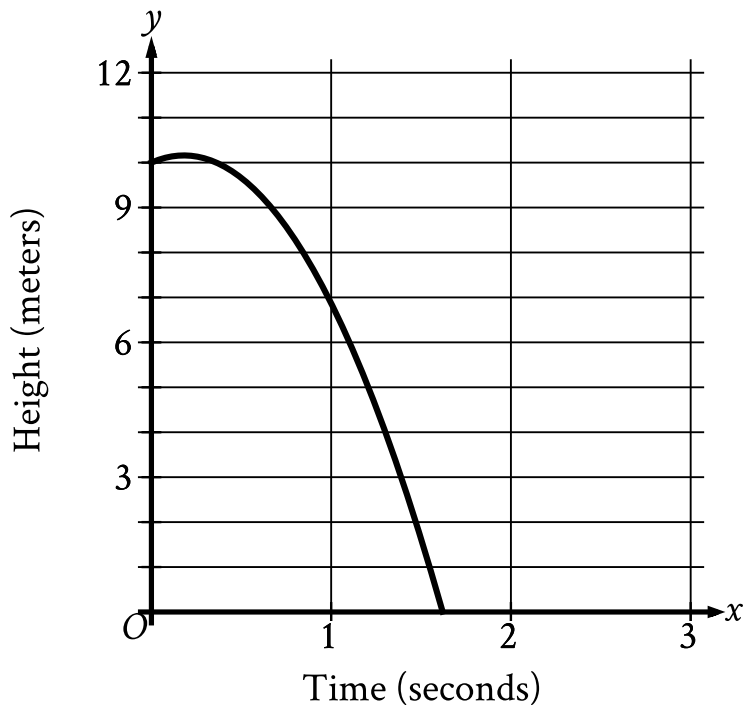
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$4a^2 + 20ab + 25b^2$$

Which of the following is a factor of the polynomial above?

- A. $a + b$
- B. $2a + 5b$
- C. $4a + 5b$
- D. $4a + 25b$



- The parabola opens downward.
- The vertex is at the approximate point (0.2 comma 10.2).
- The parabola passes through the following points:
 - (0 comma 10)
 - approximately (0.2 comma 10.2)
 - approximately (1 comma 6.9)
 - approximately (1.6 comma 0)

A competitive diver dives from a platform into the water. The graph shown gives the height above the water y , in meters, of the diver x seconds after diving from the platform. What is the best interpretation of the x -intercept of the graph?

- A. The diver reaches a maximum height above the water at 1.6 seconds.
- B. The diver hits the water at 1.6 seconds.

- C. The diver reaches a maximum height above the water at 0.2 seconds.
- D. The diver hits the water at 0.2 seconds.